THE UK UNIVERSITY INTEGRATION BEE 2024/2025

$$\int$$

ROUND TWO RELAY ROUND

Sponsored by



Jane Street

1. $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2\cos(x)}{1+e^x}\,\mathrm{d}x$

2. $\displaystyle\lim_{n\to\infty}\frac{1}{n}\int_0^1 \left(\frac{1}{n}\right)^x+\left(\frac{2}{n}\right)^x+\ldots+\left(\frac{n}{n}\right)^x\mathrm{d}x$

3. $\displaystyle\int_0^1 f(x)\,\mathrm{d}x,$

   where σ is the permutation $(3\ 1\ 4)(5\ 9\ 2\ 6)(8\ 7\ 0)$ acting on the digits $0,1,2,\ldots,9$, and $f(x)$ is the function that maps the ith decimal digit x_i of x to $\sigma(x_i)$

   Evaluate $\displaystyle\int_0^1 f(x)\,\mathrm{d}x$

4. $\displaystyle\int_0^{\pi/2} \cfrac{x^2}{2\cdot 3-x^2+\cfrac{2\cdot 3x^2}{4\cdot 5-x^2+\cfrac{4\cdot 5x^2}{6\cdot 7-x^2+\ddots}}}\mathrm{d}x$

5. $\displaystyle\int_{-\infty}^{\infty}\frac{\mathrm{d}x}{1+(x+\frac{1}{x})^2}$

6. $\displaystyle\lim_{n\to\infty} n\int_{-\frac{1}{20}}^{\frac{1}{24}}\left(1-|x|+|x|^3\right)^n\,\mathrm{d}x$

7. $\displaystyle\int_1^2 (\sqrt[x^2]{x-1})^{\mathrm{d}x}$

   where $\displaystyle\int_a^b f(x)^{\,\mathrm{d}x}$ is the product integral, defined analogously to the Riemann integral with product in place of sum, as $\displaystyle\lim_{\Delta x_i\to 0}\prod_{i=1}^{n} f(x_i)^{\Delta x_i}$ where $a=x_0<x_1<\cdots<x_n\le b$ and $\Delta x_i = x_i - x_{i-1}$.

8. $\displaystyle\int_0^{\frac{\pi}{4}}\sum_{n=2}^{\infty}\left(\frac{1}{\sqrt{n-\sec^2(x)}}-\frac{1}{\sqrt{n-\tan^2(x)}}\right)\mathrm{d}x$

9. $\displaystyle\lim_{N\to\infty}\int_0^1 (1-x)\prod_{n=0}^{N}\left(x^{2^n}+\frac{\left\{\sinh^2(x^{2^n})\right\}}{\left\{\cosh^2(x^{2^n})\right\}}\right)\,\mathrm{d}x$

10. $\displaystyle\int_{-\infty}^{\infty}\frac{e^{i\theta r}-1-i\theta\sin(r)}{|r|^{1+\alpha}}\mathrm{d}r \qquad \text{for } \alpha\in(0,1)\cup(1,2) \text{ and } \theta\in\mathbb{R},$

    where i is the imaginary unit.

11. $\displaystyle\int_0^1 \tan^{-1}\left(\frac{\tan^{-1}(\frac{x}{1-x})}{\tan^{-1}(\frac{1-x}{x})}\right)\,\mathrm{d}x$

12. $\displaystyle\int \frac{\frac{\frac{1}{\sqrt{x}}+2}{\sqrt{x+\sqrt{x}}}+4}{\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}}+8}{\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}}\,\mathrm{d}x$

13. $\int_{\frac{1}{4}}^{\frac{1}{2}} (1+\ln(x))\ln(1+x^x)\,\mathrm{d}x$

14. $\int_0^{2025} |1-|2-|3-\cdots-|2025-x|\ldots|||\,\mathrm{d}x$

15. $\int_0^1 \frac{\sin^2(\frac{\pi}{2}x)}{\sqrt{x-x^2}}\mathrm{d}x$

16. $\int_{\pi}^{3\pi} \max(\min(x,\sin(x)),\min(x,\cos(x)))\,\mathrm{d}x$

17. $\int_1^{\infty} \frac{x}{\lfloor x\rfloor^4+4}\,\mathrm{d}x$

18. $\int_{-1}^{1} \ln\left(\frac{1+x}{1-x}\right)\ln(1+e^x)\mathrm{d}x$

19. $\int_{\frac{1}{e}}^{\infty} \frac{\sqrt{\ln x+1}}{x^2}\,\mathrm{d}x$

20. $\int \frac{(x^2+x+1)(x^3-x+1)}{(x^5+1)}\mathrm{d}x$